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MS4092 COMPREHENSIVE EXAMINATION

Mechanical Design

Semester II 2022/2023

Saturday, 4th March 2023, Duration: 36 minutes

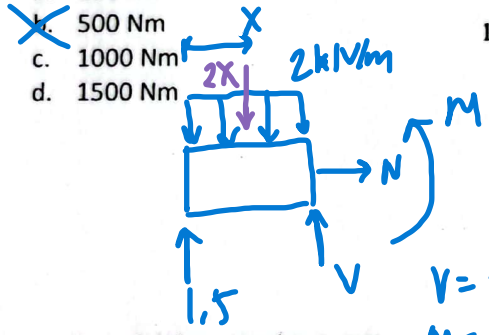
Closed Book and Closed Note

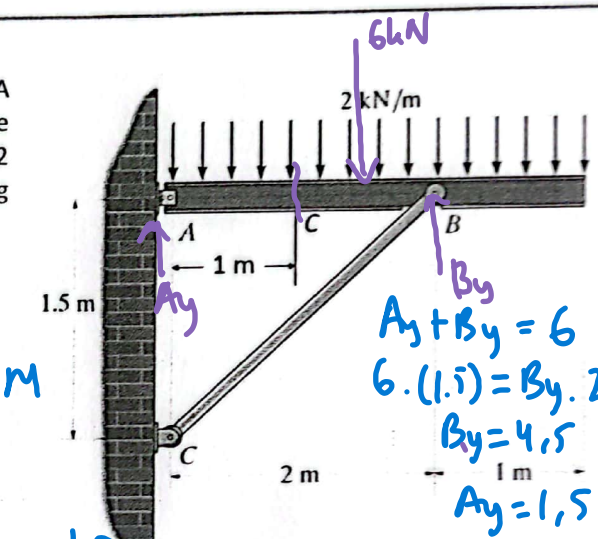
Answer all the questions on this problem sheet. State your answer by circling the letter in front of the most correct/accurate answer. Correct answers get additional scores, while wrong answers or questions that are not answered get zero score. If necessary, you may use the back page of the question sheet as 'scratch paper'. Don't forget to fill in your name, student identity number, and/or your signature at the top of EVERY page (in the space provided).

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1. A beam is supported by a pin at point A and a bar BC as shown in the figure. The beam is loaded by distributed forces 2 kN/m. What is the internal bending moment of the beam at the point C?

   a. 100 Nm
   b. 500 Nm
   c. 1000 Nm
   d. 1500 Nm





$A_y + B_y = 6$

$6.(1.5) = B_y . 2$

$B_y = 4,5$

$A_y = 1,5$

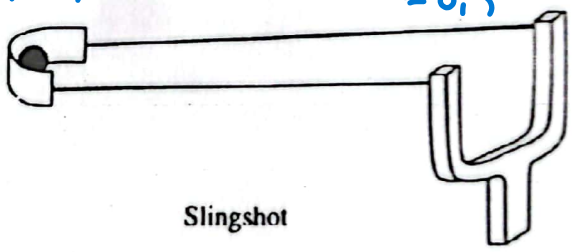
$V = -1,5 + 2x$

$M = 1,5x - x^2$

$M(1) = 1,5(1) - 1^2$

$= 0,5$

2. As a practical joke, you have put some fast-acting glue in your friend's slingshot as in figure below. Your friend puts a mass of 0.06 kg in the slingshot, pulls it back 1 m, and then releases it. If the mass travels out 1 m beyond the slingshot and then returns to strike his hand 0.5 second after the release, what must the spring constant be for the massless elastic band? Assume that the elastic band has an unstretched length equal to 0 m.



   a. 8.5 N/m
   b. 9.0 N/m
   c. 9.5 N/m
   d. 10.0 N/m

$T = 0,5\ s$

$\omega_n = \frac{2\pi}{T} = 12,57\ rad/s$

$\omega_n = \sqrt{\frac{k}{m}}$

$k = m\omega_n^2$

$= 0,06 \times (12,57)^2$

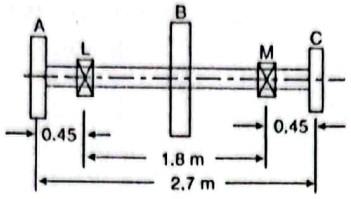
$k = 9,47 \approx 9,5\ N/m$

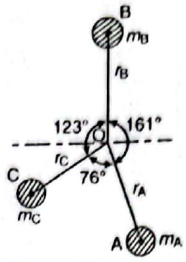
3. A shaft is supported by two bearings 1.8 *m* apart and projects 0.45 *m* beyond the bearings at each end as shown in figure below. The shaft carries three pulleys, one at each end and one at the midspan. The mass of end pulleys are m_A = 48 *kg* and m_C = 20 *kg*, with centers of gravity at r_A = 15 *mm* and r_C = 12.5 *mm*, respectively from the shaft axis. The center pulley has a mass of m_B = 56 *kg*, and its center of gravity, r_B is located 15 *mm* from the

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shaft axis. For the position shown in the Figure 2, determine the magnitude of the dynamic forces on the bearing A when the shaft rotates at 300 *rpm*.





l	*m* [kg]	*R* [mm]	*a* [m]	θ [deg]	mR sin θ	mR cos θ	mRa sin θ	mRa cos θ
A	48	15	-0.45	199	-234.41	-680.77	105.48	306.35
B	56	15	0.9	0	0.00	840.00	0.00	756.00
C	20	12.5	2.25	123	209.67	-136.16	471.75	-306.36
L			0		345.43	396.93	0	0
M			1.8		-320.69	-419.99	-577.24	-755.99
					0.00	**0.00**	**0.00**	**0.00**

a. 500 N
b. 525 N
c. 550 N
d. Not enough data

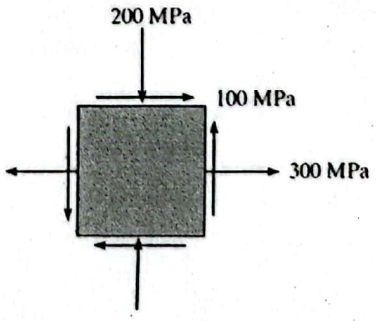
$\omega = \frac{300 \times 2\pi}{60} = 10\pi$ rad/s

$F = \sqrt{(mR\omega^2 \sin\theta)^2 + (mR\omega^2 \cos\theta)^2} = 519.32$ N

4. Based on Brown and Chandrasekaran, the design process of gearbox is the example of which class of design?
   a. Routine design
   b. Innovative design
   c. Creative design
   d. Critical design

5. Stability of a system implies that...
   a. Small changes in the system input does not result in large change in system output
   b. Small changes in the system parameters does not result in large change in system output
   c. Small changes in the initial conditions does not result in large change in system output
   d. All of the above mentioned

6. What is the principal stress of a point that has the following state of stress.



   a. 220 MPa
   b. 270 MPa
   c. 320 MPa
   d. 370 MPa

$\sigma_x = 300$ MPa

$\sigma_y = -200$ MPa

$\tau_{xy} = 100$ MPa

$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \sqrt{250^2 + 100^2}$

$R = 269.26$ MPa

$C : \left(\frac{\sigma_x + \sigma_y}{2}, 0\right)$

$(50, 0)$

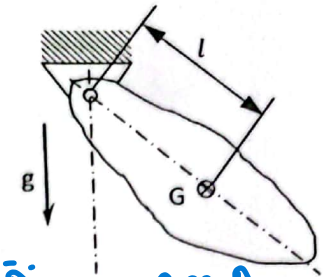
$\sigma_1 = C + R = 319.26$ MPa

$\sigma_2 = C - R = -219.26$ MPa

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7. See the pendulum at right. If G = center of mass of the pendulum, m = pendulum mass, I_G = mass moment of inertia of pendulum relative to G, I_o = mass moment of inertia of pendulum relative to pin, l = distance of pin and G, then the magnitude of natural frequency of the pendulum is about:

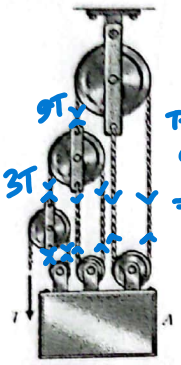


a. $\sqrt{\frac{mgl}{I_o}}$ b. $\sqrt{\frac{ml}{I_o}}$ c. $\sqrt{\frac{mgl}{I_G}}$

d. $\sqrt{\frac{ml}{I_G}}$

$I_o\ddot{\theta} + mgl\sin\theta = 0$

$\sin\theta \approx \theta$

8. 

The mass of suspended object A in the figure is m_A and the massed of the pulleys are negligible. Determine the force T necessary for the system to be in equilibrium if the gravity acceleration is g.

a. $T = \frac{1}{6}m_A g$

b. $T = \frac{1}{14}m_A g$

c. $T = \frac{1}{26}m_A g$

d. $T = \frac{1}{42}m_A g$

9T, 3T, T+T+3T+9T+9T+9T = 26T

9. Find the critical value of the gain K for stability in the following open-loop transfer function below.

$$G(s)H(s) = \frac{K}{s(s+1)(2s+1)}$$

a. 1

b. 1.5

c. 2

d. 2.5

$2s^3 + 3s^2 + s + K = 0$

$\frac{3-2K}{3} = 0 \rightarrow K = 1.5$

s^3	2	1	0
s^2	3	K	0
s^1	$\frac{3-2K}{3}$	0	
s^0	K		

10. What term is used for the maximum stress at which material fail on a specified number of cycle?

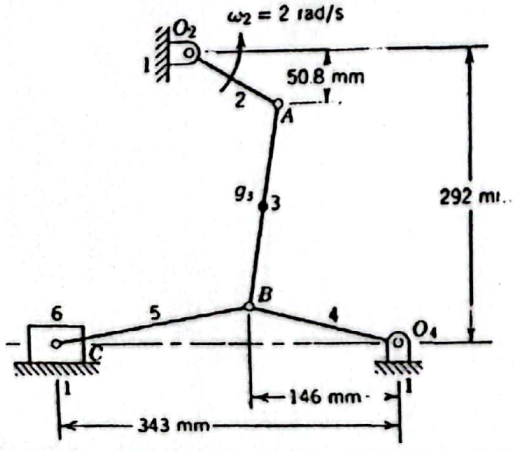
a. **Fatigue strength**

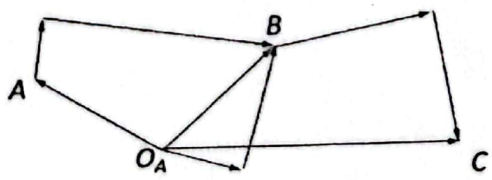
b. Fatigue life

c. Ultimate tensile strength

d. Endurance limit

11. In the six-bar mechanism below, it is known that the center of mass of link 3, g_3, is located at the middle of the link. It is also provided acceleration polygon of the mechanism.



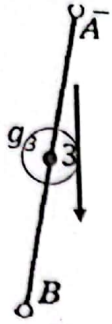


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Which following inertia force location and direction for link 3 is correct?

a.



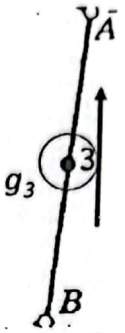
c.



b.



d.



12. Two mating spur gears have 40 and 120 teeth respectively. The pinion rotates at 1200 rpm and transmit torque of 20 Nm. The torque transmitted by the gear is...

a. 6.6 Nm
b. 20 Nm
c. 40 Nm
d. 60 Nm

$F_1 = F_2$

$\frac{T_1}{d_1} = \frac{T_2}{d_2}$

$T_2 = \frac{20}{40} \cdot 120 = 60$ Nm

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